Introduction To Computers

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1. Computer Software

Computer software is a **set of instructions or programs** that tell the computer how to perform tasks.

intangible — you can't touch it,

but it **controls** the operation of the **computer hardware**.

2. Computer Software

enables users to **interact with the computer**

& perform specific tasks such as **typing a document**, **browsing the internet**, or **playing a game**.

Major Types of Software

1) System Software
2) Application Software
3) Programming Software

1) System Software

System software **manages and controls computer hardware** so that application software can function.

Examples . . .

a) Operating System (OS)
b) Device Drivers
c) Utility Programs
d) Firmware

a) Operating System

An Operating System (OS) is system software that **manages computer hardware**, **software resources**, and **provides services for computer programs**.

It acts as a bridge between the user and the computer hardware.

(e.g., Windows, macOS, Linux, Android)

a) Operating System

Without an operating system, a computer would be useless —

users wouldn't be able to run applications or even see anything on the screen.

i) Key Functions of Operating System

a) User Interface (UI)

- Provides a way for users to **interact** with the computer (e.g., Windows GUI or Linux CLI).

b) File Management

- Organizes, stores, retrieves, and secures data files on storage devices.

i) Key Functions of Operating System

c) Memory Management

- Allocates and deallocates memory space to applications and processes.

d) Process Management

- Controls how programs are executed (task scheduling, multitasking).

i) Key Functions of Operating System

e) Device Management

- Manages input/output devices using drivers (e.g., printers, USB, keyboard).

f) Security and Access Control

- Protects data and system access using passwords, permissions, and encryption.

i) Key Functions of Operating System

g) System Performance Monitoring

- Tracks system health, CPU usage, and background processes.

b) Device Drivers

Helps OS communicate with hardware

(e.g., printer driver, display driver)

c) Utility Programs

Perform specific maintenance tasks

(e.g., antivirus, disk cleanup tools)

d) Firmware

Embedded software in hardware

(e.g., BIOS in motherboard)

2) Application Software

Application software helps users **perform specific tasks.**

Examples . . .

Microsoft Word (Word processing — creating/editing documents)

Microsoft Excel (Spreadsheet calculations and data analysis)

2) Application Software

… Examples . . .

Google Chrome (Web browsing)

Adobe Photoshop (Image editing)

Zoom / Skype (Video communication)

Games (Entertainment and learning (e.g., Minecraft, Chess software))

3) Programming Software

Used by developers to **create**, **debug**, and **maintain other software and applications**.

Examples …

Code Editors

- Writing source code (e.g., Notepad++, Visual Studio Code)

3) Programming Software

… Examples …

Compilers

- Converts source code into executable form (e.g., GCC, Turbo C)

Interpreters

- Executes code line-by-line (e.g., Python Interpreter)

3) Programming Software

… Examples …

Debuggers

- Helps detect and fix programming errors

IDEs (Integrated Development Environments)

- Combines editor, compiler, debugger (e.g., Eclipse, IntelliJ, Visual Studio)

Hardware vs Software

Lets' as a summary, we discuss the differences between

Hardware and Software

Hardware vs Software

Feature – **Definition**

(Hardware) - The physical components of a computer system

(Software) - The set of instructions or programs that run on hardware.

Hardware vs Software

Feature – **Tangibility**

(Hardware) - **Tangible** – you can touch and see it.

(Software) - **Intangible** – cannot be touched or seen physically.

Hardware vs Software

Feature – **Function**

(Hardware) - Performs actual tasks like input, output, storage.

(Software) - Tells hardware what to do by providing instructions.

Hardware vs Software

Feature – **Failure Effect**

(Hardware) - If damaged, it might need **replacement** or **repair**.

(Software) - If corrupted, it can be **reinstalled** or **updated**.

Hardware vs Software

Feature – **Interaction**

(Hardware) - Works **with** software to complete tasks

(Software) - Needs hardware to be **executed** or **installed**.

Hardware vs Software

Feature – **Change Frequency**

(Hardware) - Changes **less frequently** – may last for years.

(Software) - Changes **frequently** – new versions and updates released..

Hardware vs Software

Feature – **Storage**

(Hardware) - Stored **physically** inside the computer

(Software) - Stored **digitally** on storage devices.

Hardware vs Software

Feature – **Examples**

(Hardware) - Keyboard, Monitor, CPU, Hard Disk, Mouse.

(Software) - MS Word, Windows, Google Chrome, Photoshop.